Reset Early, Reset Often, Saturate the SAT Core

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Abstract

We improve PySATCoP performance by resetting the connection tableau often. This causes more and more varied SAT statements to be added to the SAT prover, which causes it to saturate faster. The resulting prover solves problems faster and in significantly fewer steps than SATCoP and other state-of-the-art tableau provers. [Add actual numbers here.]

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1 Introduction

[Write an introduction that enhances and complements the abstract.]

2 Preliminaries

2.1 Connection Calculus and Tableau Solving Methods

Connection tableaux and model elimination give a goal-directed alternative to the saturation style used by provers such as Vampire[16]. Instead of deriving clauses by proving repeatedly derive clauses from a global clause set until a contradiction is found or the search space saturates[1, 19], connection calculi instead try to close a tableau by repeatedly connecting the goal literals to complementary literals, either on the current path or in a fresh copy of an input clause[7, 11].

The standard clausal connection tableau has three proof-building inferences[7, 12].

Start Choose an input clause and make its literals open goals.

Extension Open a branch on a clause containing a complementary literal, and add the remaining literals as new subgoals. This might require a substitution of variables with predicates.

Reduction Closes the current goal against a complementary literal on the branch. Like the previous step, this might require adding a substitution.

The calculus is sound and complete with backtracking and iterative deepening, but it is not confluent: a locally valid connection can bind variables in a way that prevents later branches from closing[14].

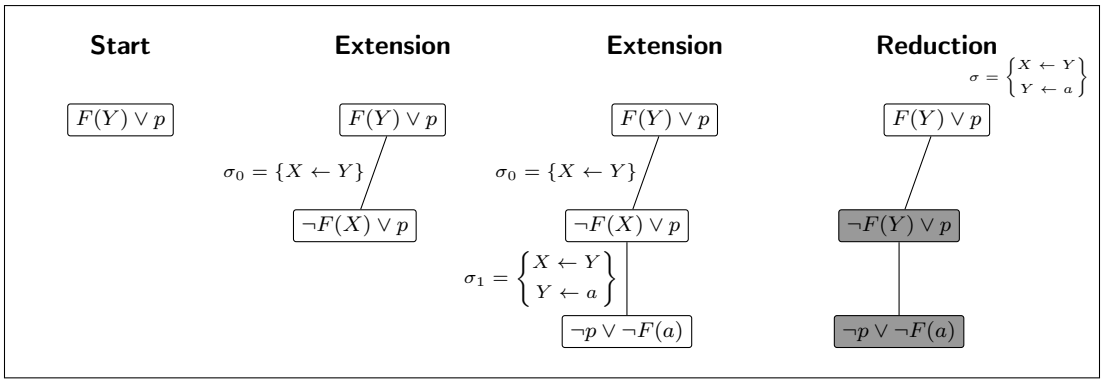


Figure 1: The three proof-building steps of a clausal connection tableau, taken from the tableau closure of SYN052-1.p[13]. The two Extension steps add an extra clause to the tableau; the Reduction step closes the nodes that were proven unsat. A full solution can be found in Figure 4.

This control problem is what made the "lean" Prolog line of tableau provers influential. leanTAP showed that a competitive free-variable tableau prover could be implemented very compactly in Prolog[2]. leanCoP adapted this style to connection tableaux and became the base of a large family of provers for classical and non-classical logics[10, 9]. The important engineering lesson from leanCoP is that a tiny change to backtracking can dominate performance. The `cut` option commits after successful Reduction and Extension steps using Prolog's `!`, removing many alternative local continuations[8]. Since this restricted backtracking is incomplete, leanCoP combines it with iterative-deepening schedules such as `comp(7)`, where the prover searches with cut up to a depth bound and then restarts without the cut restriction if necessary[8, 9]. In a bounded run, these choices are not merely optimisations; they decide which parts of the proof space are even explored.

Several later systems can be read as attempts to make this local control less crude. meanCoP separates cut on reductions from cut on extensions and introduces exclusive cuts, which preserve more alternatives than the inclusive cuts used by leanCoP[4]. Rawson, Eisenhofer, and Kovacs take the opposite route: instead of deliberately throwing away alternatives, they learn constraints explaining why a search state is stuck, thereby reducing repeated backtracking while preserving completeness at a fixed iterative-deepening level[14]. Functional connection provers and the mlCoP/monteCoP line reorder proof steps using learned information or Monte Carlo Tree Search[5]. rlCoP and plCoP make this more explicit by treating proof search as a learning-guided control process over leanCoP-style actions[6, 22, 23]. The common theme is the one we use below: connection-tableau proving is naturally viewed as a sequence of small control decisions over proof actions, cuts, backtracking, and restarts.

2.2 SATCoP and Similar Methods

The other line of work relevant to this paper adds a second, ground-level proof search underneath the tableau search. SAT and SMT solvers have long been used to support first-order reasoning, for example in finite model finding, instance generation, and splitting architectures such as AVATAR[15, 18]. For a connection prover, the useful observation is especially simple. Every clause that appears in a tableau is an instance of an input clause. If we ground these tableau clauses and collect them in a propositional SAT solver, then a SAT refutation of the collected ground clauses is a sound refutation of the original first-order problem[15].

Rawson and Reger implement this idea in SATCoP, a clausal connection-tableau prover based on leanCoP[15]. During backtracking search, clauses present in the tableau are heuristically grounded and added to an incremental SAT solver. If the SAT solver reports that the accumulated ground clauses are unsatisfiable, proof search can stop even if the first-order tableau itself has not been closed[15]. The remaining tableau variables must be mapped to ground terms before a clause is sent to SAT. Their implementation follows a Schulz-style heuristic: use the most frequent constant in the conjecture, and fall back to a fresh constant if no conjecture constant is available[15, 18, § 4.2].

SATCoP also uses the SAT solver for control. First, the accumulated ground clauses are kept across tableau backtracking, so failed first-order search leaves useful propositional information behind[15]. Second, iterative deepening is modified: when a bounded search pass discovers new ground clauses, the prover can restart at the same depth rather than immediately increasing the depth bound[15, § 5]. Third, the current SAT model gives model-based lemmata: if a ground literal on the current path is already false in the SAT model, refuting that branch would not eliminate the current model and the tableau branch can be skipped[15, § 6]. In a later note included in the same paper, Rawson and Reger also describe a literal-selection refinement: prefer extension choices whose selected literal is true in the current SAT model[15, § 9.1].

This paper keeps the same basic soundness argument as SATCoP but changes the emphasis. leanCoP, meanCoP, and learned connection provers mainly try to close the tableau faster. SATCoP shows that the tableau can instead be used to generate a growing stream of useful ground clauses. Our strategy, SATResetCoP, pushes this second view further: when the current tableau search stops producing new SAT clauses, we reset it early in order to generate different ground instances and saturate the SAT core faster. Thus the relevant historical thread is not just "tableau prover plus SAT solver", but a gradual movement from tableau closure as the only objective towards explicit control of what information the tableau search should produce.

3 Research Vehicle

3.1 SATResetCoP Specification

We propose implementing a new tableau prover, based on SATCoP, but whose tableau prover attempts to saturate the SAT prover rather than to close the tableau.

Successive experimentation resulted in a simple hypothesis: aggressively resetting the tree when the SAT solver stops growing results in a larger quantity and variety of SAT statements added to the solver, which results in reaching a SAT unsatisfiability proof faster.

SATResetCoP uses the same substitution and SAT prover unsatisfiability technique as PySATCoP. However, instead of following leanCoP's tableau guidance strategy, it prioritises resetting the tableau often. The policy is sketched below; the ordering of forward actions follows the SAT-model literal selection idea of Rawson and Reger[15, § 9.1].

```
def solve_tableau(start_clauses):
  clauses := ground(start_clauses)

  max_depth := 1
  tableau := empty_tableau(max_depth)

  while solve_sat(clauses) != UNSAT:
    Actions := legal_actions(tableau)  // may include Backtrack

    model_rank(lit) :=
      0 if ground(lit) is true in SAT(clauses)
      1 if ground(lit) is unassigned in SAT(clauses)
      2 if ground(lit) is false in SAT(clauses)
      3 if lit = Backtrack

    a := argmin(a in Actions, model_rank(selected_literal_or_backtrack(a)))

    if a != Backtrack:
      step(a)
      clauses.add(ground(tableau))
      continue

    if ground(tableau) adds a new clause:
      tableau := empty_tableau(max_depth)
    else:
      tableau := empty_tableau(max_depth + 1)
      max_depth := max_depth + 1
```

Listing 1: Sketch of the SATResetCoP control policy.

Just like PySATCoP, this approach is sound. While it is less complete than PySATCoP and other leanCoP variants, the speed at which it proves unsatisfiability of first-order problems can result in a larger amount of problems solved in a bounded amount of time or steps. It might be useful to incorporate SATResetCoP into the initial steps of a more complete schedule.

3.2 Implementation Details

The grounded clauses used by PySATCoP and SATResetCoP come from the clauses currently present in the tableau, after applying the tableau substitution accumulated so far. Ground terms are kept as they are; only tableau variables that remain unassigned need to be replaced by a ground term.

As explained in Section 2.2, Rawson and Reger follow the technique by Schulz[18] to use the most frequent constant in the conjecture, with a fresh constant as a fallback when the conjecture contains no constants[15]. SATResetCoP always uses this fallback-style choice, mapping every remaining tableau variable to the same fresh artificial constant. This gives a grounding scheme that is simpler than SATCoP, while keeping the proof sound: the SAT solver still only receives ground instances of clauses already present in the tableau.
[Randomisation of statements.]

3.3 Python Implementation

As a testbed for out experiments we use the Connections Prober by Rømming et. al[17], which provides an implementation of leanCoP-style provers.

The connections prover makes it easy to implement different strategies to guide the tableau proof. This makes implementing the guidance of SATResetCoP simple and its code easy to read and evaluate.

The source code for this implementation is present in https://github.com/mfixman/resetsatcop, and is released under the GNU Lesser General Public Licence[3].

3.4 TPTP Benchmarking

Along with out implementation of SATResetCoP, PyCoP allows us to reimplement PyLeanCoP and PySATCoP in its own backend. This provides a fairer baseline for comparison.

We run SATResetCoP along with our versions of these two connection tableau strategies, along with a strategy that chooses the first possible action as control, on the 8344 CNF problems and 9091 FOF problems of all 50 categories [Categories?] containing at least one of these problems from the Thousands of Problems for Theorem Provers (TPTP) library[20].

The library provides a varied set of problems that's a standard way to prove first-order provers. It also provides declared statuses for these problems, which are a useful sanity check for false unsatisfiability results.

4 Experimental Evaluation

4.1 Experiment Design

The connections prover provides with PyCoP, a simple prover that chooses the first legal non-backtrack action at each point and resets the tree with an increased maximum height saturated.

As a second control policy we use our implementation of leanCoP[8], PyLeanCoP. This is a Python version of the leanCoP Prolog implementation, with the two recommended predicates used in the first step of the schedule by Otten[8, § 5.5].

`cut` Choose one choice from each extension and reduction and backtrack afterwards, in a similar way to to Prolog's `!` operator.

`comp(7)` The prover does iterative deepening of height 1 at a time when the tree is saturated until reaching maximum height 7. Afterwards limits on height are restricted.

These kinds of simple policies tend to be very effective in tableau provers[9].

The TPTP library contains 8344 CNF problems and 9091 FOF problems separated in 50 categories[20]. Each problem/model pair is run for a maximum of 90 seconds and 1 million steps, on a single core of the University of Cambridge CSD3 Cluster with [add CPU information] and 3370 MiB RAM per core[21].

4.2 Results

4.2.1 Individual Results

SATResetCoP considerably improves the amount of problems solved by both PySATCoP_0 and PySATCoP under the 90 second time limit, as shown in Table 2. This is mostly caused by the SAT solved reaching unsatisfiability in fewer time and steps than in previous models, as seen in Figures 2a and 2b.

The 1 000 000 step limit is rarely reached, and most unsolved problems hit the 90 second limit instead.

Model	Explanation	Cut	UNSAT Proof	SAT Guidance	Reset on Backtrack
PyCoP PyLeanCoP	Simple control policy leanCoP with initial set of predicates.	✓			
PySATCoP$_0$	Based on the initial implementation of SATCoP[15].	✓	✓		
PySATCoP	Improvements on SATCoP[15, § 9.1].	✓	✓	✓	
SATResetCoP	Novel model, specified in Section 3.1.	✓	✓	✓	✓

Table 1: The four control models and the test model used in this experiment.

Model	CNF	FOF	Total
PyCoP	205	587	792
PyLeanCoP	701	1155	1856
PySATCoP$_0$	787	1042	1829
PySATCoP	1027	1396	2423
SATResetCoP	1227	1634	2861

Table 2: Amount of CNF and FOF problems solved within the time constraint by each model.

4.2.2 Comparative Results

The SAT guidance in SATResetCoP causes it to solve several problems that are unsolved by either models which do not have the same SAT saturation prover, and even by PySATCoP$_0$ and PySATCoP, which do not have similar efficient guidance. A comparative result can be seen in Table 3.

SATResetCoP is specially strong in problems which can be solved by small unsat cores. For example, Figure 4 shows the solution to the TPTP problem SYN052-1[13].

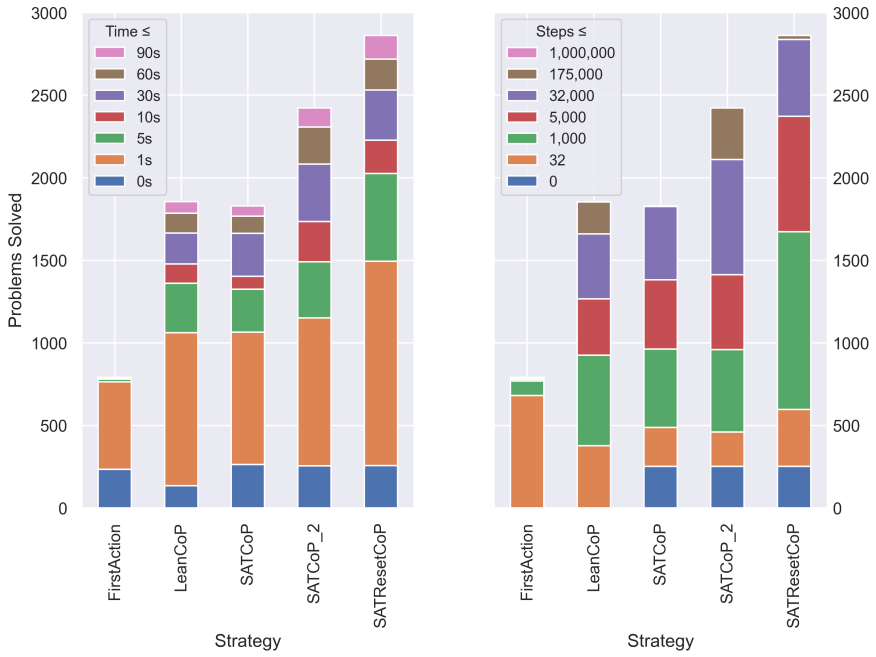
Figure 6 shows the amount of problems solved one of the models and not a previous one.

The size of the improvement between SATResetCoP and the next best model is notable.

[Add more analysis here, including size of unsat cores]

Model	CNF	FOF
PyCoP	205	587
PyLeanCoP	502	693
PySATCoP$_0$	184	129
PySATCoP	188	189
SATResetCoP	213	228

Table 3: Problems solved by each model and not solved by any preceding model in the table. [What's the point of separating CNF and FOF here? Maybe finding interesting TPTP types would be better.]



(a) Problems solved, stacked by time in seconds.

(b) Problem solved, stacked by amount of steps. SAT models can solve a problem in 0 steps if the initial grounding produces is SAT-unsatisfiable.

Figure 2: Results of running the models on all problems solved in a 90 second and 1 million step limit.

[Time comparison / violinplot of different TPTP problem types for SATResetCoP]

Figure 3: Time taken by SATResetCoP to solve each problem within each TPTP problem type. We can probably find some conclusion in here.

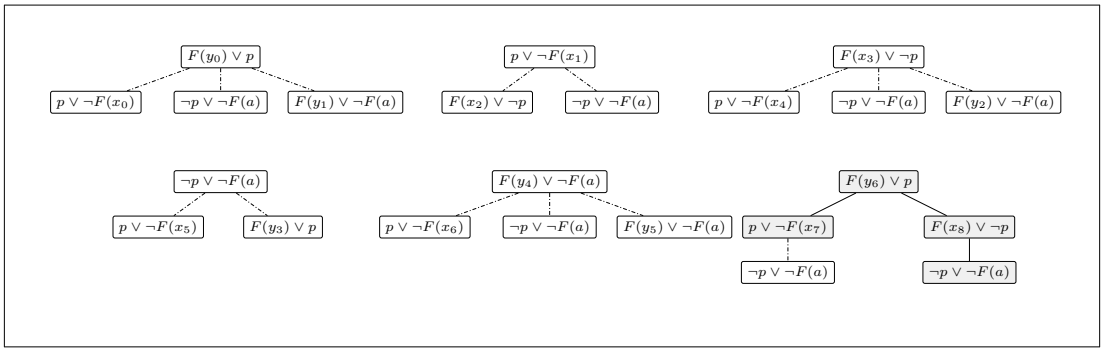
4.2.3 Soundness Verification: Problem Satisfiability

The problems in the TPTP library have a helpful specification of the satisfiability of different types of problems[20]. This is useful as a sanity-check: a sound solver should never "prove" unsatisfiability on a satisfiable problem, and their authors should be wary when proving unsatisfiability on an open or unknown problem. Note that this is not a way to prove soundness of the solver, but to check that it does not exhibit false unsatisfiability results on problems declared satisfiable or countersatisfiable.

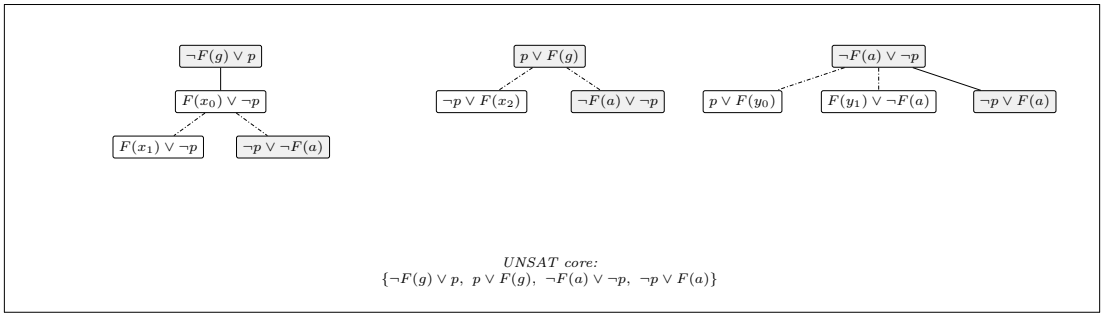
We make a second run of SATResetCoP with a longer 10-minute limit, whose results can be seen in Table 4. These results do not show any negative results.

$$p \vee \neg F(X) \qquad F(X) \vee \neg p$$
$$F(Y) \vee p \qquad \neg p \vee \neg F(a) \qquad F(Y) \vee \neg F(a)$$

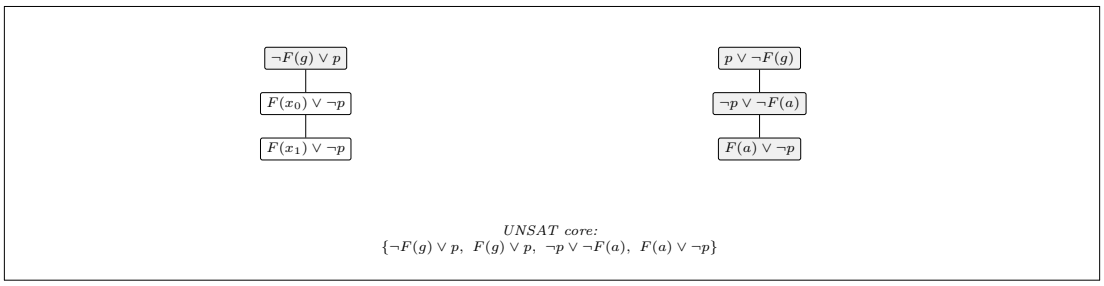
(a) Problem SYN052-1.p[13], adapted from https://tptp.org/TPTP/Problems/SYN/SYN052-1.p.



(b) Solution found by PyLeanCoP in 43 steps.



(c) Solution found by SATCoP in 34 steps.

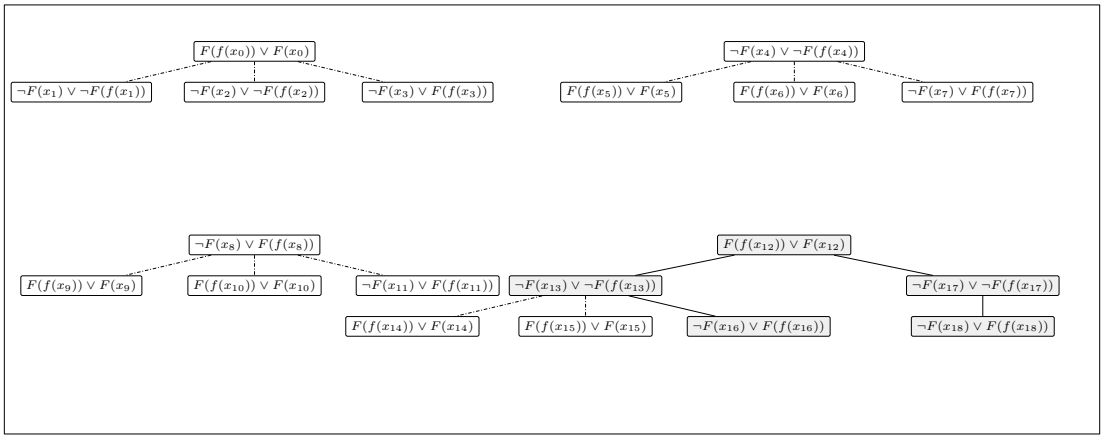


(d) Solution found by SATResetCoP in 11 steps.

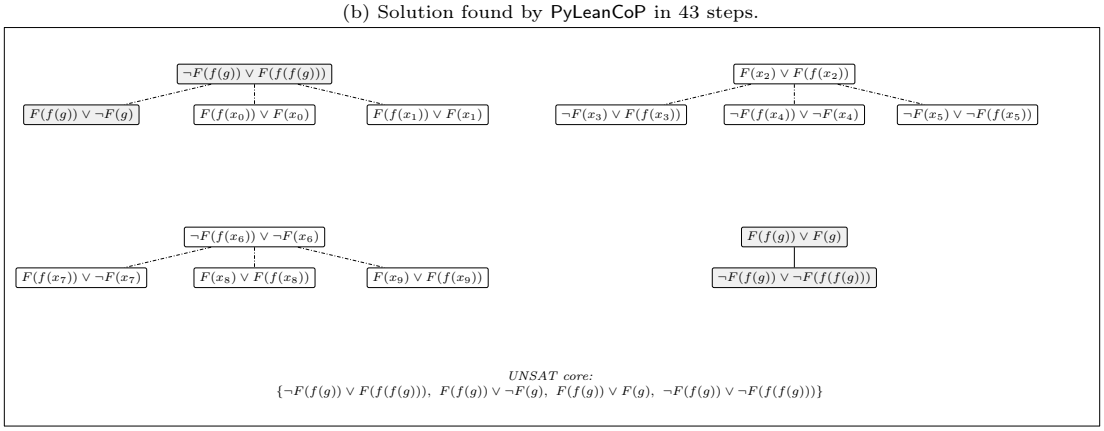
Figure 4: The unsatisfiability proof to the TPTP problem SYN052-1[13] coming from three provers: PyLeanCoP, PySATCoP, and SATResetCoP.

$$\neg F(X) \vee \neg F(f(x)) \qquad F(f(X)) \vee F(x)$$
$$\neg F(X) \vee F(f(X))$$

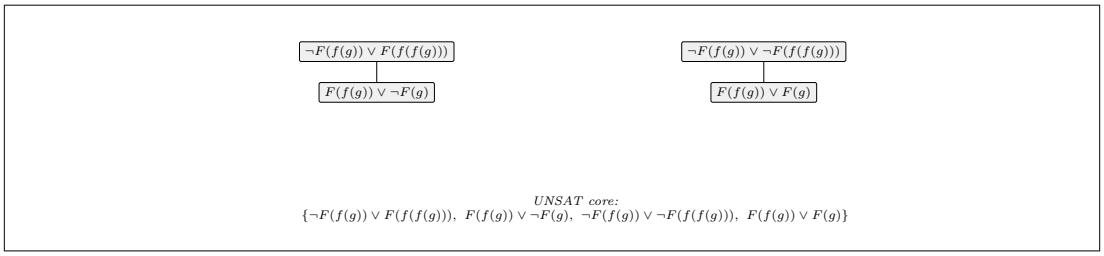
(a) Problem SYN081-1.p[13], adapted from https://tptp.org/TPTP/Problems/SYN/SYN081-1.p.



(b) Solution found by PyLeanCoP in 43 steps.



(c) Solution found by SATCoP in 34 steps.



(d) Solution found by SATResetCoP in 11 steps.

Figure 5: The unsatisfiability proof to the TPTP problem SYN081-1[13] coming from three provers: PyLeanCoP, PySATCoP, and SATResetCoP.

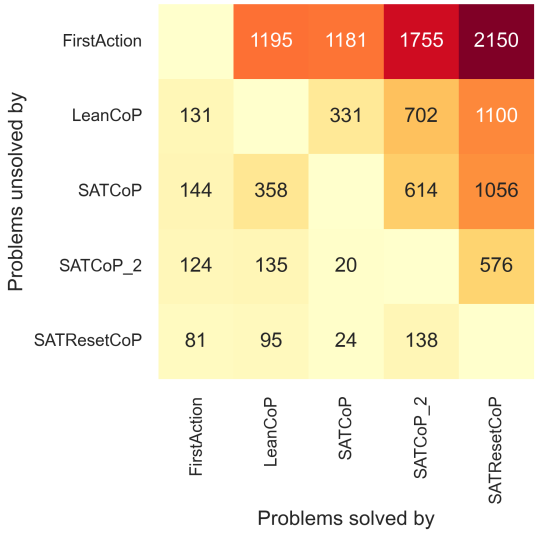


Figure 6: Pairwise comparison of model results: the cell in column y and row x shows the amount of problems solved by the model on y, but not the model in x. For example, SATResetCoP solved 1056 TPTP problems that PySATCoP$_0$ did not. [Remake this figure with new names.]

	FOF		CNF	
Declared status	Unsolved	Solved	Unsolved	Solved
Satisfiable	350	0	1094	0
CounterSatisfiable	732	0	0	0
Unknown	94	0	745	0
Open	4	0	28	0
Unsatisfiable	100	72	4496	1433
ContradictoryAxioms	204	39	0	0
Theorem	4426	1823	0	0

Table 4: Amount of problems solved and unsolved by SATResetCoP, grouped by TPTP declared status. No satisfiable or countersatisfiable problem was incorrectly proven to be unsatisfiable.

5 Future Work

6 Acknowledgements

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